

Lecture 7: Time Series
Tasks and Dynamical Systems
Chaos, Embedding and Signal Characteristics

Deep Learning in Classical and Quantum Physics

Marcin Płodzień

Atomic Optics Department
Institute of Theoretical Physics
Jagiellonian University in Kraków

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Time Series and the Forecasting Problem



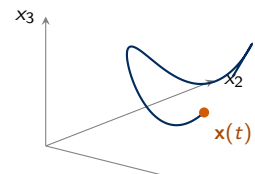
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State of a System

State and equations of motion

$$\mathbf{x}(t) = (x_1(t), \ldots, x_d(t)), \qquad \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \text{i.e. } \dot{x}_i = F_i(x_1, \ldots, x_d)$$

x_i: variables that change in time, together they fix the state; $\mathbf{F}$: deterministic, time-independent, smooth, in general nonlinear



state $\mathbf{x}(t) \in \mathbb{R}^d$, a point moving in phase space

- pendulum: angle, angular velocity; laser: field, polarisation, inversion
- parameters (viscosity, heating): constants inside $\mathbf{F}$, not state variables
- $\mathbf{x}(t)$ moves along a trajectory in phase space

The Law of Motion F

$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$: everything that drives the system

F contains all effects acting on the state: forces, couplings between the components, friction and dissipation, external driving, delays, and all parameters.

F known well

pendulum, springs, electrical circuits, planetary orbits

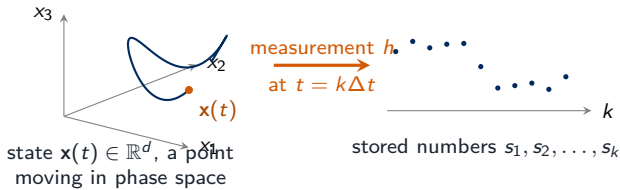
F known only approximately

cells, gene regulation, ecosystems, climate, the brain

missing terms, uncertain parameters, unknown couplings

- where **F** is imperfect, the measured record is the main source of knowledge about the system

Measured Time Series



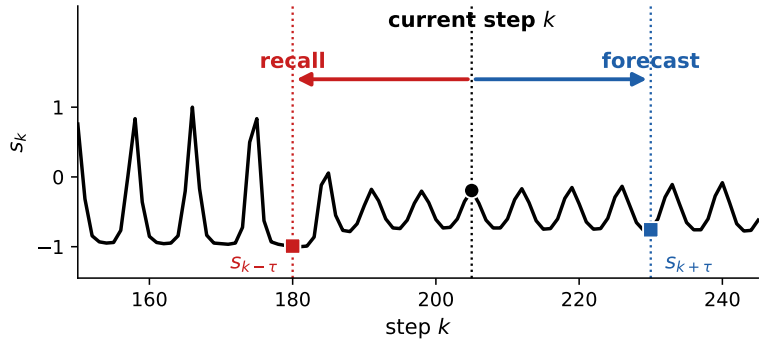
Time series

$$s_k = h(\mathbf{x}(k\Delta t)) + \eta_k, \quad k = 1, \ldots, N$$

h: measurement function; η_k: measurement noise; Δt: sampling interval

- one stored number per step, a function of all components of $\mathbf{x}(k\Delta t)$
- $\mathbf{x}$ itself is never accessible
- η_k only corrupts the record; dynamical noise, $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) + \boldsymbol{\xi}(t)$, changes the future

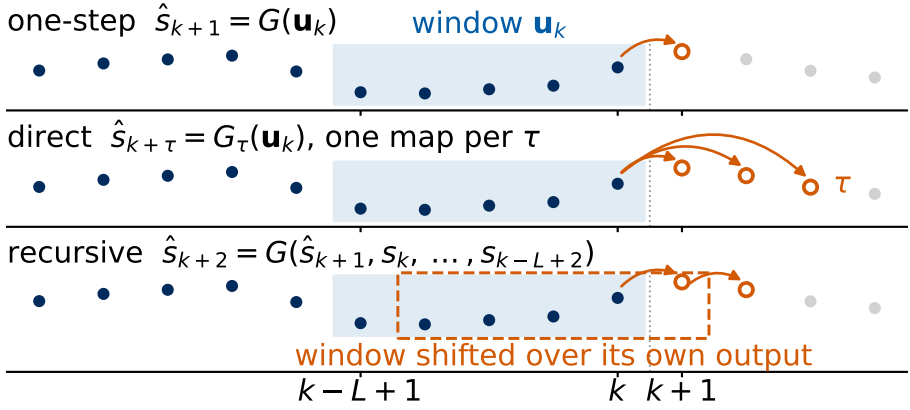
Recall and Forecast Tasks



Task: infer future values $s_{k+\tau}$ or past values $s_{k-\tau}$ from the stored $s_1, \ldots, s_k$ alone, without access to $\mathbf{x}$.

- **recall**: target $y_k = s_{k-\tau}$, a value τ steps in the past, which tests memory
- **forecast**: target $y_k = s_{k+\tau}$, a value τ steps ahead, τ is the horizon

Forecast, Horizon and Window



dark dots: stored values s_j; grey dots: future values; blue box: window $\mathbf{u}_k$ of the last L values; orange circles: forecasts $\hat{s}$; dashed orange box: window shifted over its own forecast; model $\hat{G}_\tau$, written G in the drawing

Forecast Map and Window Prescription

Forecast at horizon τ

$$\hat{s}_{k+\tau} = \hat{G}_\tau(\mathbf{u}_k), \qquad \mathbf{u}_k = (s_k, \ldots, s_{k-L+1})$$

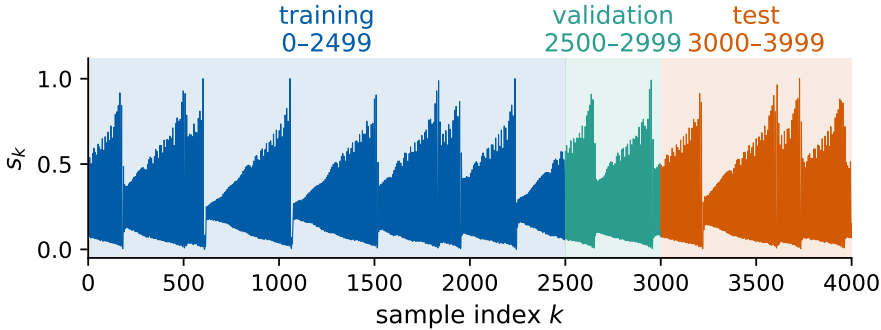
$\hat{G}_\tau$: our model for the value τ steps ahead (recall: back), hat: estimated from data; $\mathbf{u}_k$: last L values, a stand-in for $\mathbf{x}$

Aim

1. a window prescription: spacing κ and number m of past values that fix $\mathbf{x}(k\Delta t)$
2. a model $\hat{G}_\tau$ learned from the record

J. D. Farmer, J. J. Sidorowich, Phys. Rev. Lett. 59, 845 (1987)

Training and Testing in Time Order



Every signal: $N = 4000$ samples, split in time order.

- fit on the training block, choose settings on the validation block, report on the test block
- never shuffle: random windows would put test values into the training inputs

Pearson Correlation and Capacity

Pearson correlation and capacity

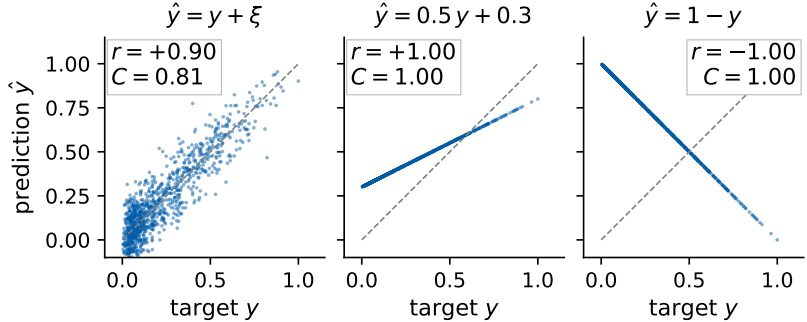
$$r(\tau) = \frac{\sum_i (\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2 \sum_i (y_i - \bar{y})^2}}, \qquad C(\tau) = r(\tau)^2, \qquad C_{\mathrm{tot}} = \sum_{\tau=1}^{50} C(\tau)$$

$\hat{y}_i$, y_i: forecast and target at test step i; bars: means over the test block

- $C(\tau) \in [0, 1]$: fraction of the target variance explained by the best straight line through the forecast
- C_{tot}: one number for all horizons $\tau = 1, \ldots, 50$

J. Dambre *et al.*, Sci. Rep. 2, 514 (2012)

Measure of Forecast Quality



forecast $\hat{y} = \hat{s}_{k+\tau}$ against target $y = s_{k+\tau}$, three constructed forecasts; dashed: $\hat{y} = y$

- ideal: all points on the diagonal $\hat{y} = y$
- C: fit to a straight line; noise lowers it, scale and offset do not
- $\hat{y} = 1 - y$ also has $C = 1$: check the sign of r

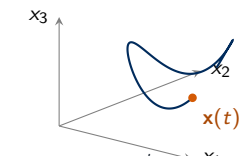
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Dynamical Systems



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Phase Space, Flows and Maps



state $\mathbf{x}(t) \in \mathbb{R}^d$, a point moving in phase space

Phase space

The set of all possible states $\mathbf{x} \in \mathbb{R}^d$: each point is one complete state, the motion is a trajectory through it.

Flow and map

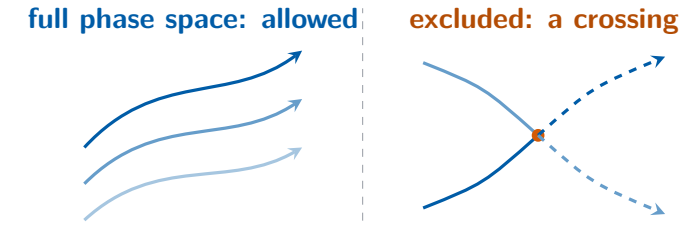
$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$: flow, continuous time

$\mathbf{x}_{n+1} = \mathbf{f}(\mathbf{x}_n)$: map, discrete steps

- flow: e.g. a pendulum, Lorenz-63; map: e.g. the logistic map $x_{n+1} = 4x_n(1 - x_n)$
- a flow sampled every Δt is a map, $\mathbf{x}_{k+1} = \varphi_{\Delta t}(\mathbf{x}_k)$: what a stored record sees
- **chaos**: bounded, never repeating, nearby states separate exponentially

S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., CRC Press (2018), doi:10.1201/9780429492563

Trajectories Never Cross



blue curves: trajectories of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$; orange dot: a crossing point; dashed: its two continuations

- $\mathbf{F}$ continuously differentiable: one and only one solution through every state
- a crossing would give one state two futures, so trajectories never cross
- a measured projection can cross itself: a sign of missing state variables

Flow Map

Trajectory and flow map of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$

$$\text{trajectory:} \quad t \mapsto \mathbf{x}(t; \mathbf{x}_0), \quad \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}),\ \mathbf{x}(0) = \mathbf{x}_0$$

$$\text{flow map:} \quad \varphi_t : \mathbb{R}^d \to \mathbb{R}^d, \quad \mathbf{x}_0 \mapsto \varphi_t(\mathbf{x}_0) \coloneqq \mathbf{x}(t; \mathbf{x}_0)$$

φ_t moves every point of phase space along its own trajectory for the time t

Properties, for smooth $\mathbf{F}$ independent of time

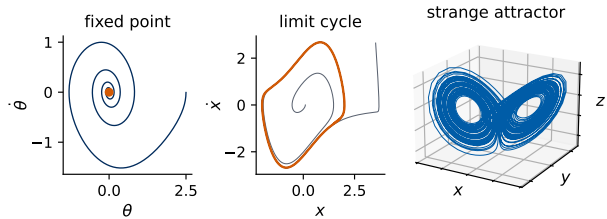
$$\varphi_0(\mathbf{x}_0) = \mathbf{x}_0 \qquad \text{initial condition}$$

$$\varphi_{t+s} = \varphi_t \circ \varphi_s \qquad \text{composition}$$

$$\varphi_{-t} = \varphi_t^{-1} \qquad \text{inverse}$$

- example $\dot{x} = ax$: $\varphi_t(x_0) = e^{at} x_0$; sampled record: $\mathbf{x}_{k+1} = \varphi_{\Delta t}(\mathbf{x}_k)$

Attractor



Attractor

The set A in phase space on which the motion ends up after the transient, for all starts in a region, its **basin**; once on A, a trajectory stays on A.

- pendulum: rest; Van der Pol $\ddot{x} = (1 - x^2)\dot{x} - x$: two starts (grey), one periodic orbit (orange)
- **strange** (Lorenz-63): orbit never closes, nearby orbits separate; fractal set, non-integer dimension (2.06), from stretching and folding of a contracting flow

Phase-Space Volume

Volume of a set of states

$$A_t := \varphi_t(A_0) = \{\varphi_t(\mathbf{x}_0) : \mathbf{x}_0 \in A_0\} \qquad \text{states at time } t \text{ of all starts in } A_0$$

$$V(t) := \int_{A_t} d^d y \qquad \mathbf{y}\text{: point of } A_t$$

$$(M_t)_{ij}(\mathbf{x}_0) := \frac{\partial \varphi_{t,i}(\mathbf{x}_0)}{\partial x_{0,j}} \qquad \text{Jacobian of the flow map, } M_0 = I$$

$$\mathbf{y} = \varphi_t(\mathbf{x}_0) \;\Rightarrow\; d^d y = \det M_t(\mathbf{x}_0)\, d^d x_0 \qquad \text{change of variables}$$

$$V(t) = \int_{A_0} \det M_t(\mathbf{x}_0)\; d^d x_0 \qquad \text{integral over the starts}$$

A_0: set of initial states, e.g. a small ball; each $\mathbf{y} \in A_t$ comes from exactly one $\mathbf{x}_0 \in A_0$

- $\det M_t(\mathbf{x}_0) = \delta V(t)/\delta V(0)$: volume ratio of a small element around $\mathbf{x}_0$

Liouville's Theorem

Liouville's theorem

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \int_{A_t} \nabla \cdot \mathbf{F}\, d^d y \quad \text{for every } A_0, \qquad \nabla \cdot \mathbf{F} = \sum_{i=1}^{d} \frac{\partial F_i}{\partial x_i}$$

Hamiltonian flow: $\nabla \cdot \mathbf{F} = 0$, so the phase-space volume is conserved

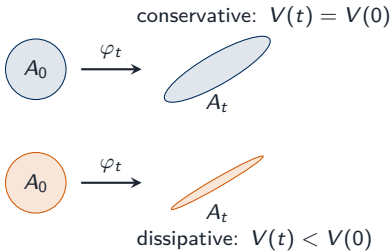
Small set A_0

$$\frac{\mathrm{d}V}{\mathrm{d}t} \approx (\nabla \cdot \mathbf{F})(\mathbf{x}(t))\; V(t)$$

A_t small: $\nabla \cdot \mathbf{F}$ nearly constant on A_t, equal to its value at the centre $\mathbf{x}(t)$, and comes out of the integral

- $\nabla \cdot \mathbf{F}$ is a rate: $\frac{1}{\delta V}\frac{\mathrm{d}\,\delta V}{\mathrm{d}t} = (\nabla \cdot \mathbf{F})(\mathbf{x}(t))$ for a small element, so $\det M_t(\mathbf{x}_0) = \exp\big(\int_0^t (\nabla \cdot \mathbf{F})(\mathbf{x}(s))\,\mathrm{d}s\big)$
- constant $\nabla \cdot \mathbf{F}$: $V(t) = V(0)\, e^{(\nabla \cdot \mathbf{F})\, t}$ for any A_0

Conservative and Dissipative Flows



- Hamiltonian: $\nabla \cdot \mathbf{F} = 0$, the volume is conserved, $V(t) = V(0)$; A_t can stretch and fold, never shrink
- dissipative: $\nabla \cdot \mathbf{F} < 0$, the volume shrinks and the motion settles on an attractor of lower dimension
- Lorenz-63: $\nabla \cdot \mathbf{F} = -41/3$ at every state, $V(t) = V(0)\, e^{-41t/3}$ for any A_0

Lorenz-63 as Truncated Convection

Fluid layer heated from below (Rayleigh–Bénard), Boussinesq approximation

$$\partial_t \mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\tfrac{1}{\varrho_0}\nabla\mathcal{P} + \nu\nabla^2\mathbf{v} + g_{\mathrm{gr}}\,\alpha\,(\Theta-\Theta_0)\,\hat{\mathbf{e}}_Z, \qquad \nabla\cdot\mathbf{v} = 0$$

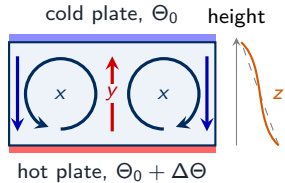
$$\partial_t \Theta + (\mathbf{v}\cdot\nabla)\Theta = \chi\nabla^2\Theta$$

$\mathbf{v}$, Θ: velocity, temperature; Θ_0: cold-plate temperature; $\mathcal{P}$: pressure; ϱ_0: reference density; ν: kinematic viscosity; χ: thermal diffusivity; α: thermal expansion; g_{gr}: gravity; $\hat{\mathbf{e}}_Z$: upward unit vector

- state: two fields, infinitely many dimensions
- Saltzman, Lorenz: 2D rolls, stream function ϖ, free-slip plates, Fourier modes
- keep three modes x, y, z: Lorenz-63, qualitative only, far above the onset of convection

B. Saltzman, J. Atmos. Sci. 19, 329 (1962); E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963)

Lorenz-63 Model



Fluid heated from below, $\mathbf{x} = (x, y, z)$:

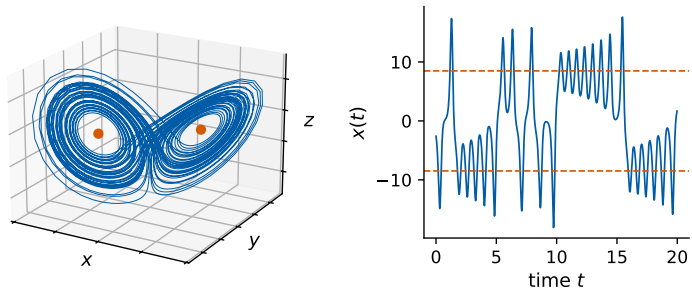
$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} \sigma(y - x) \\ x(\rho - z) - y \\ xy - \beta z \end{pmatrix}$$

x: speed of the rolls, sign: sense of rotation; y: temperature difference, rising (red) against sinking (blue) fluid; z: bending of the profile (orange) away from conduction (dashed)

- $\sigma = \nu/\chi = 10$: Prandtl number
- $\rho = \mathrm{Ra}/\mathrm{Ra}_c = 28$: heating, Rayleigh number over its value at onset
- $\beta = 4/(1 + a^2) = 8/3$: roll geometry, aspect ratio a of the rolls, $a^2 = 1/2$

E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963)

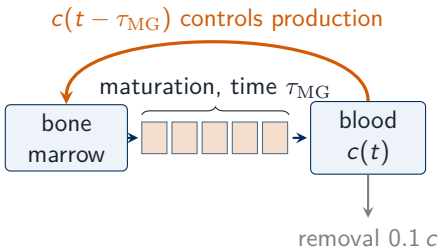
Lorenz-63 Strange Attractor



$\rho = 28$; left: trajectory in (x, y, z), dots: fixed points of the two steady rolls; right: $x(t)$, dashed: x at the fixed points

- $\rho < 1$ rest; $1 < \rho < 24.74$ steady rolls; $\rho = 28$ the rolls flip irregularly
- $\nabla \cdot \mathbf{F} = -(\sigma + 1 + \beta) = -41/3$: $V(t) = V(0)\, e^{-41t/3}$; stored signal $s_k = x(k\Delta t)$, $\Delta t = 0.02$

Mackey–Glass Model



Production follows the concentration τ_{MG} ago:

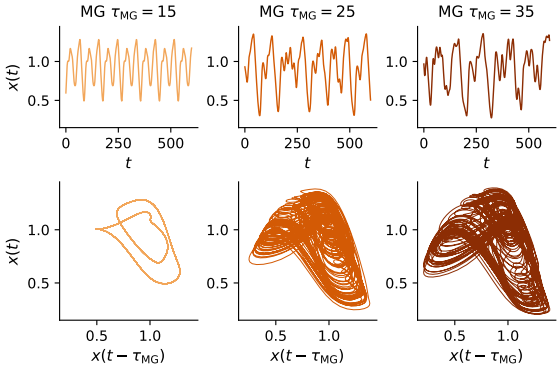
$$\dot{c} = \underbrace{\frac{0.2\,c(t-\tau_{\mathrm{MG}})}{1+c(t-\tau_{\mathrm{MG}})^{10}}}_{\text{production}} - \underbrace{0.1\,c}_{\text{removal}}$$

c: blood-cell concentration; removal rate 0.1 per time unit

- continuing from t needs the whole history on $[t-\tau_{\mathrm{MG}}, t]$: infinite-dimensional phase space
- larger delay: steady state, then periodic, then irregular motion

M. C. Mackey, L. Glass, Science 197, 287 (1977), doi:10.1126/science.267326

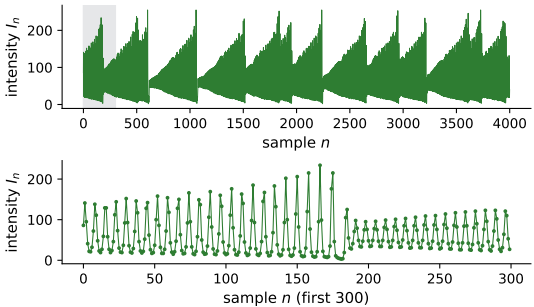
Mackey–Glass Solutions



c, written x on the axes; top: $c(t)$ for $\tau_{\mathrm{MG}} = 15, 25, 35$; bottom: $c(t)$ against $c(t - \tau_{\mathrm{MG}})$

- $\tau_{\mathrm{MG}} = 15$: limit cycle, a closed line; $\tau_{\mathrm{MG}} = 25, 35$: chaotic, the orbit never closes and fills a band

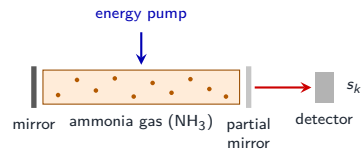
Santa Fe Laser



intensity s_k, written I_n in the drawing, against sample k, written n; top: 4000 samples used; bottom: first 300 samples

U. Hübner *et al.*, Phys. Rev. A 40, 6354 (1989)

Far-infrared ammonia laser: measured 8-bit intensity (Santa Fe competition).



- three variables: field E, polarisation P, inversion W
- pulses grow and collapse

Laser Equations in Lorenz Form

Maxwell–Bloch equations of a single-mode laser

$$\dot{E} = -\gamma_c(E - gP), \qquad \dot{P} = -\gamma_\perp(P - EW), \qquad \dot{W} = -\gamma_\parallel(W - W_0 + EP)$$

E: cavity field; P: polarisation; W: population inversion, W_0 without light; γ: decay rates; g: coupling

Rescaling, time $t' = \gamma_\perp t$

$$x = \sqrt{\beta}\,E, \quad y = \sqrt{\beta}\,gP, \quad z = g(W_0 - W), \qquad \sigma = \frac{\gamma_c}{\gamma_\perp}, \quad \beta = \frac{\gamma_\parallel}{\gamma_\perp}, \quad \rho = gW_0$$

gives $x' = \sigma(y - x)$, $y' = x(\rho - z) - y$, $z' = xy - \beta z$, the Lorenz-63 equations

- pump $\leftrightarrow$ heating ρ; unstable lasing needs $\sigma > \beta + 1$, a bad cavity
- detector sees $E^2 \propto x^2$: spiral growth on one lobe, switch to the other $\rightarrow$ growth and collapse

H. Haken, Phys. Lett. A 53, 77 (1975); C. O. Weiss, J. Brock, Phys. Rev. Lett. 57, 2804 (1986)

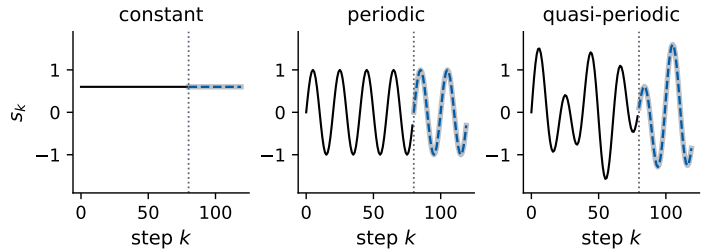
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Regular and Chaotic Signals



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Regular Signals



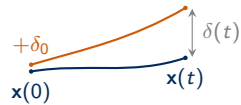
Past (black) up to $k = 80$, true continuation (grey) and forecast (dashed blue).

- **constant**: the last value, $\hat{s}_{k+\tau} = s_k$ (persistence)
- **periodic** with period T_p: the value one period earlier, $\hat{s}_{k+\tau} = s_{k+\tau-T_\mathrm{p}}$
- **quasi-periodic**, two sines with an irrational frequency ratio: never repeats, yet obeys an exact linear recurrence $s_k = \sum_{j=1}^{4} a_j s_{k-j}$, fitted on the past and iterated

All three forecasts are exact to round-off, these are trivial tasks.

Propagation of Small Errors

Equations of motion $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$; second trajectory $\tilde{\mathbf{x}}(t) = \mathbf{x}(t) + \delta(t)$, $\delta(0) = \delta_0$.



Linearisation

$$\dot{\delta} = \mathbf{F}(\mathbf{x}+\delta) - \mathbf{F}(\mathbf{x}) \approx J(\mathbf{x}(t))\,\delta, \quad J_{ij} = \frac{\partial F_i}{\partial x_j} \qquad \text{first order: linear in } \delta$$

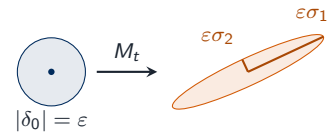
$$\delta(t) = M_t\,\delta_0, \quad (M_t)_{ij} = \frac{\partial x_i(t)}{\partial x_j(0)} \qquad \text{change of } x_i(t) \text{ per change of } x_j(0)$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(M_t)_{ij} = \sum_l J_{il}(\mathbf{x}(t))\,(M_t)_{lj}, \quad M_0 = I \qquad \text{same linear equation, identity at } t = 0$$

- M_t: Jacobian of the flow map, as for the volume; carries a small error from 0 to t

Growth Factors of the Error

The growth of a small error depends on its direction. The eigenproblem below gives the directions and their growth factors.



Growth factors

$$|\delta(t)|^2 = \delta(t)^\top \delta(t) = \delta_0^\top M_t^\top M_t\, \delta_0 \qquad \text{squared length of the error}$$

$$\tilde{M}_t \coloneqq M_t^\top M_t \qquad \text{symmetric } d \times d \text{ matrix}$$

$$\tilde{M}_t\, \mathbf{v}_i = \sigma_i^2\, \mathbf{v}_i, \quad \sigma_1 \geq \cdots \geq \sigma_d \geq 0 \qquad \text{eigenproblem, } \mathbf{v}_i \text{ orthonormal}$$

$$\delta_0 = \varepsilon \mathbf{v}_i \;\Rightarrow\; |\delta(t)| = \varepsilon\, \sigma_i \qquad \sigma_i\text{: growth factor along } \mathbf{v}_i$$

- a ball of errors becomes an ellipsoid with semi-axes $\varepsilon\sigma_1, \ldots, \varepsilon\sigma_d$
- the state stays bounded, only the error grows

Lyapunov Spectrum

Lyapunov exponents

$$\tilde{M}_t \, \mathbf{v}_i = \sigma_i^2(t) \, \mathbf{v}_i, \qquad \tilde{M}_t = M_t^\top M_t, \qquad \lambda_i = \lim_{t\to\infty} \frac{1}{t} \ln \sigma_i(t), \qquad i = 1, \ldots, d$$

$\sigma_i^2(t)$: eigenvalues of $\tilde{M}_t = M_t^\top M_t$, $\sigma_1 \geq \cdots \geq \sigma_d$; $\sigma_i(t)$: growth factor of an error along $\mathbf{v}_i$

- λ_i: growth rate of the i-th axis, $\sigma_i(t) \sim e^{\lambda_i t}$; $\lambda_1 > 0$: chaos
- flow, bounded trajectory away from fixed points: one $\lambda_i = 0$, the direction along the orbit
- $\lambda_1 + \cdots + \lambda_d = \langle \nabla \cdot \mathbf{F} \rangle$: the volume growth rate, < 0 for a dissipative system

V. I. Oseledets, Trans. Moscow Math. Soc. 19, 197 (1968); G. Benettin *et al.*, Meccanica 15, 9 (1980)

Sum of the Lyapunov Exponents

Sum rule

$$\lambda_1 + \cdots + \lambda_d = \langle \nabla \cdot \mathbf{F} \rangle := \lim_{t\to\infty} \frac{1}{t} \int_0^t (\nabla \cdot \mathbf{F})(\mathbf{x}(s))\, \mathrm{d}s$$

$\lambda_i = \lim_{t\to\infty} \frac{1}{t} \ln \sigma_i(t)$: Lyapunov exponents; $\langle \nabla \cdot \mathbf{F} \rangle$: time average along the trajectory

$$|\det M_t| = \sigma_1(t) \cdots \sigma_d(t) \qquad \det \tilde{M}_t = (\det M_t)^2 = \textstyle\prod_i \sigma_i^2$$

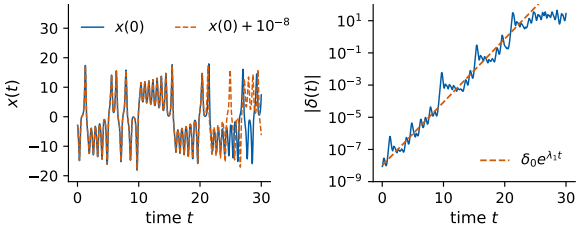
$$\frac{\mathrm{d}}{\mathrm{d}t} \det M_t = (\nabla \cdot \mathbf{F})(\mathbf{x}(t)) \det M_t, \quad \det M_0 = 1 \qquad \text{Jacobi's formula}$$

$$\Rightarrow \det M_t(\mathbf{x}_0) = \exp\Big(\int_0^t (\nabla \cdot \mathbf{F})(\mathbf{x}(s))\, \mathrm{d}s \Big) \qquad \dot{u} = a(t)\, u,\ u(0) = 1$$

$$\textstyle\sum_i \lambda_i = \lim_{t\to\infty} \frac{1}{t} \ln(\sigma_1 \cdots \sigma_d) = \langle \nabla \cdot \mathbf{F} \rangle \qquad \text{insert } |\det M_t|$$

- dissipative: $\langle \nabla \cdot \mathbf{F} \rangle < 0$; Lorenz-63: $\nabla \cdot \mathbf{F} = -41/3$, $\lambda_1 + \lambda_2 + \lambda_3 = -13.667$

Sensitivity to Initial Conditions



E. N. Lorenz, J. Atmos. Sci. 20, 130 (1963)

Lorenz-63 from two states 10^{-8} apart; left $x(t)$ of both, right their distance $|\delta(t)|$.

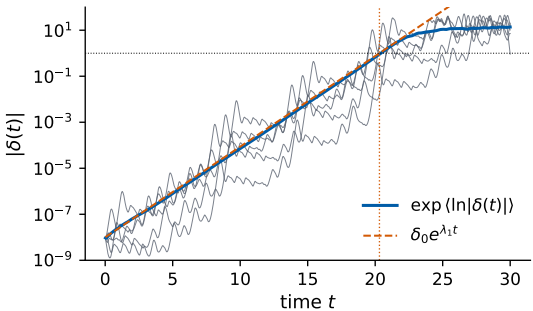
Largest Lyapunov exponent

$$|\delta(t)| \approx |\delta_0|\, e^{\lambda_1 t}, \qquad \lambda_1 = \lim_{t\to\infty} \lim_{|\delta_0|\to 0} \frac{1}{t} \ln \frac{|\delta(t)|}{|\delta_0|}$$

$\delta(t)$: difference of the two states; $T_\lambda = 1/\lambda_1$: Lyapunov time

The distance exceeds 1 at $t \approx 19.5$; slope of $\ln|\delta|$ 0.93, average $\lambda_1 = 0.907$.

Fundamental Limit of the Prediction Horizon



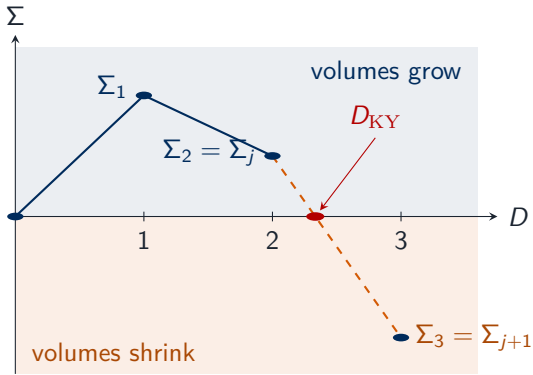
$|\delta(t)|$ of five pairs (grey) and $\exp\langle \ln|\delta|\rangle$ over 200 pairs (blue).

An initial error δ_0 grows until it reaches a tolerance Δ,

$$T_{\text{pred}} = \frac{1}{\lambda_1} \ln \frac{\Delta}{\delta_0}$$

- Lorenz, $|\delta_0| = 10^{-8}$: $\Delta = 1$ at $t = 20.4$, formula 20.3
- start 1000 times more precise, $\delta_0 \to \delta_0/1000$: only $\ln(1000)\, T_\lambda \approx 6.9\, T_\lambda$ longer
- chaos ($\lambda_1 > 0$): finite horizon even for the exact equations; a learned model stays below T_{pred}

Dimension Where Volumes Stop Growing



sketch, $j = 2$: dots: partial sums Σ_ℓ; dashed: the line $\Sigma(D)$; axes: dimension D, growth rate Σ

- $\Sigma_\ell = \lambda_1 + \cdots + \lambda_\ell$: growth rate of ℓ-dimensional volumes, only at whole ℓ
- j: last ℓ with $\Sigma_\ell \geq 0$; straight line $\Sigma(D)$ between j and $j+1$
- D_{KY}: where the line crosses zero; below it volumes grow, above it they shrink

Kaplan–Yorke Dimension

Kaplan–Yorke dimension

$$\Sigma_\ell = \lambda_1 + \cdots + \lambda_\ell \qquad \text{sum of the } \ell \text{ largest exponents}$$

$$j = \max\{\ell : \Sigma_\ell \geq 0\} \qquad \text{last } \ell \text{ with } \Sigma_\ell \geq 0$$

$$\Sigma(D) \coloneqq \Sigma_j + (D - j)\,\lambda_{j+1}, \quad j \leq D \leq j+1 \qquad \text{straight line from } \Sigma_j \text{ to } \Sigma_{j+1}$$

$$\Sigma(D_{\mathrm{KY}}) = 0 \;\Rightarrow\; D_{\mathrm{KY}} = j + \frac{\Sigma_j}{|\lambda_{j+1}|} \qquad \text{where the line crosses zero}$$

- ℓ-volumes grow as $e^{\Sigma_\ell t}$; fixed point 0, limit cycle 1, chaos non-integer
- estimate from the spectrum; no general equality to the box dimension D_B
- Kaplan–Yorke conjecture, unproven in general: D_{KY} = information dimension

J. L. Kaplan, J. A. Yorke, Lect. Notes Math. 730, 204 (1979)

Kaplan–Yorke Dimension of Lorenz-63

$\ell = 1$: length grows
$\Sigma_1 > 0$



$\ell = 2$: area does not shrink
$\Sigma_2 \geq 0$



$\ell = 3$: volume shrinks
$\Sigma_3 < 0$

ℓ	λ_ℓ (per time unit)	$\Sigma_\ell = \lambda_1 + \cdots + \lambda_\ell$
1	0.907	0.907
2	0.0001	0.907
3	-14.573	-13.667

- $j = 2$: the largest dimension whose pieces do not shrink
- $D_{\mathrm{KY}} = 2 + \Sigma_2/|\lambda_3| = 2.06$: an attractor a little more than a surface
- $\Sigma_3 = \nabla \cdot \mathbf{F} = -41/3$: the volume contraction of the flow

Fractional Dimension of Lorenz-63

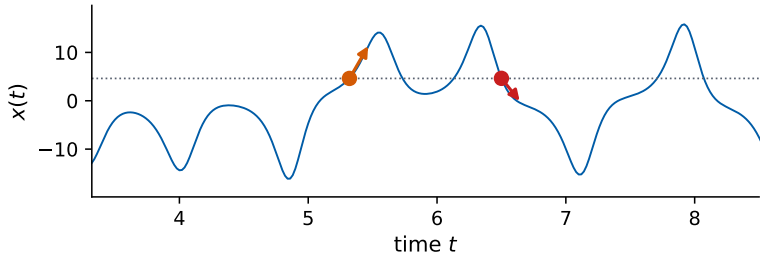
D_{KY}: how many dimensions the attractor occupies, from how the flow stretches and squeezes small regions.



sketch: a point, a curve, a surface, and a cross-section of the Lorenz attractor drawn as thin layers (illustration, not computed positions)

- volumes shrink ($\Sigma_3 < 0$), areas do not ($\Sigma_2 \geq 0$): dimension between 2 and 3
- the bounded, stretched sheet is folded again and again: the extra 0.06 is the fine layering
- forecasting: about $\lceil D_{\mathrm{KY}} \rceil = 3$ independent numbers locate a state on the attractor

One Value Does Not Fix the State



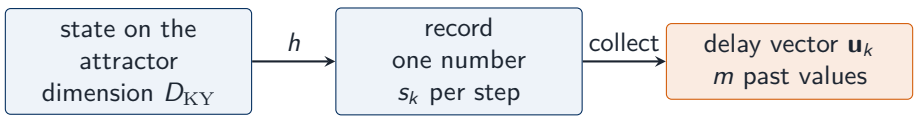
Lorenz-63 $x(t)$; two moments with the same value (dotted), one rising, one falling

- state: a point $\mathbf{x} = (x, y, z)$ in phase space; the equations of motion fix its future
- we measure $s_k = x$, the projection of the point on one axis
- same x, different y, z: different points, different futures; a forecast of s_{k+1} needs past values $s_k, s_{k-\kappa}, \ldots$

Dimension and the Number of Measurements

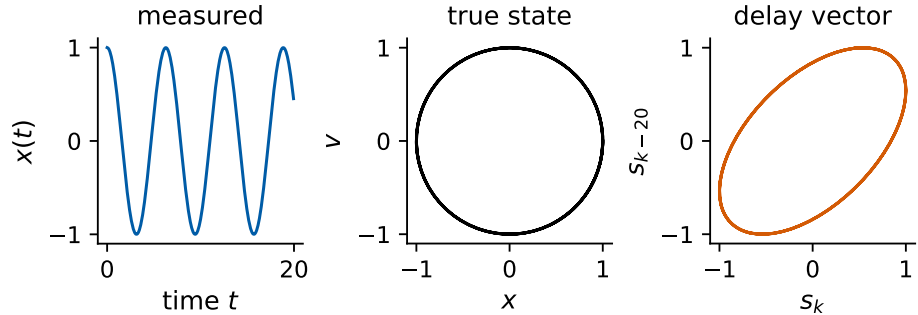
Aim

number m and spacing κ of past values such that the delay vector $\mathbf{u}_k = (s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa})$ fixes the state $\mathbf{x}_k$, and with it every future value $s_{k+\tau}$



- a point on the attractor: about $\lceil D_{\mathrm{KY}} \rceil$ numbers, 3 for Lorenz-63
- one measurement per step is one number; past values add the missing ones, because the dynamics wrote y, z into the later x
- at least $\lceil D_{\mathrm{KY}} \rceil$ values; a sufficient number follows from the embedding theorems

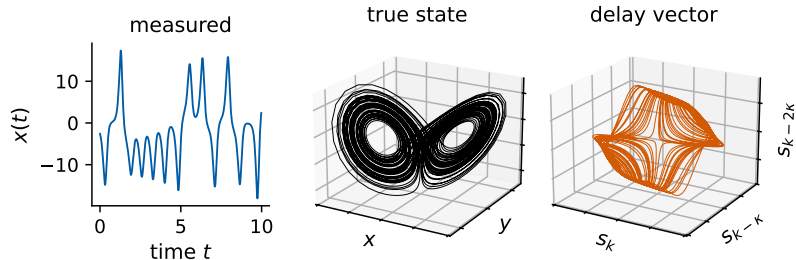
Phase Space from One Coordinate



harmonic oscillator $x(t) = \cos t$, $\Delta t = 0.05$; left: $x(t)$; middle: true state (x, v); right: pairs (s_k, s_{k-20}), $s_k = x(k\Delta t)$, lag 20 samples

- $v_k \approx (x_k - x_{k-1})/\Delta t$: the hidden velocity is written into the past values
- the pairs trace a deformed copy of the same closed curve

Lorenz-63 from One Coordinate



left: measured $x(t)$; middle: true state (x, y, z); right: delay vectors $(s_k, s_{k-\kappa}, s_{k-2\kappa})$, $s_k = x(k\Delta t)$, $\kappa = 10$ samples

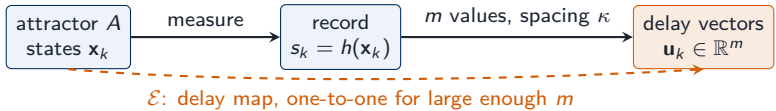
True state and delay vector

$\mathbf{x}_k = (x, y, z)(k\Delta t)$: three quantities at one time

$\mathbf{u}_k = (s_k, s_{k-\kappa}, s_{k-2\kappa})$: one quantity at three times

$\mathbf{u}_k \neq \mathbf{x}_k$, yet both lobes reappear in deformed coordinates

Delay Map



Delay map $\mathcal{E}$: state $\to$ vector in $\mathbb{R}^m$

$$\mathcal{E}(\mathbf{x}) \coloneqq \big(h(\mathbf{x}),\ h(\varphi_{-\kappa\Delta t}(\mathbf{x})),\ \ldots,\ h(\varphi_{-(m-1)\kappa\Delta t}(\mathbf{x}))\big) \qquad \text{measure at } 0, -\kappa, -2\kappa, \ldots \text{ steps}$$

$$\mathbf{x}_{k-\kappa} = \varphi_{-\kappa\Delta t}(\mathbf{x}_k) \ \Rightarrow\ h(\varphi_{-\kappa\Delta t}(\mathbf{x}_k)) = h(\mathbf{x}_{k-\kappa}) = s_{k-\kappa} \qquad \text{flow map backwards}$$

$$\mathcal{E}(\mathbf{x}_k) = \big(s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa}\big) = \mathbf{u}_k \qquad \text{from the record}$$

h: measurement function, state $\to$ one stored number; κ: delay; m: dimension

- $\mathbf{u}_k$: numbers computed from the record; equal to $\mathcal{E}$ of the hidden true state $\mathbf{x}_k$
- $\mathcal{E}$, h, φ are never needed explicitly: the theorem is about $\mathcal{E}$, the data give $\mathbf{u}_k$

Takens Embedding Theorem

Embedding theorems (Takens; Sauer, Yorke, Casdagli)

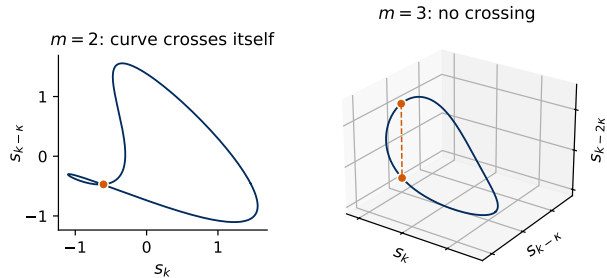
1. A on a smooth manifold of dimension D_{M}: $m \geq 2D_{\mathrm{M}} + 1$ gives an embedding, for generic flow map and h
2. A of box-counting dimension D_B: $m > 2D_B$ gives a one-to-one $\mathcal{E}$ on A, for almost every smooth h, mild conditions on low-period orbits

A: bounded closed set carrying the motion; embedding: one-to-one with continuous inverse on $\mathcal{E}(A)$; $D_B = \lim_{\varepsilon \to 0} \ln N_\varepsilon / \ln(1/\varepsilon)$, N_ε: boxes of side ε covering A

- one-to-one: $\mathbf{u}_k$ fixes $\mathbf{x}_k$; topology kept, distances and angles distorted
- dynamics carries over: $\mathbf{u}_{k+1} = \mathcal{E} \circ \varphi_{\Delta t} \circ \mathcal{E}^{-1}(\mathbf{u}_k)$

F. Takens, Lect. Notes Math. 898, 366 (1981); T. Sauer, J. A. Yorke, M. Casdagli, J. Stat. Phys. 65, 579 (1991)

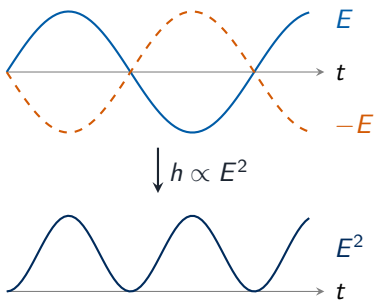
Embedding Dimension $m > 2D_B$



$s(t) = \cos t + 0.6\cos(2t + 0.7)$, delay a quarter period: closed curve, $D_B = 1$; left $(s_k, s_{k-\kappa})$, right $(s_k, s_{k-\kappa}, s_{k-2\kappa})$; orange: two different states

- $m = 2$: two states share a delay vector; $m = 3 > 2D_B$: one state per delay vector
- two pieces of dimension D in $\mathbb{R}^m$ meet in general only if $2D \geq m$
- D_{KY}: practical proxy for D_B, no guarantee; heuristic $\lceil D_{\mathrm{KY}} \rceil \leq m_{\mathsf{min}} \leq \lfloor 2D_{\mathrm{KY}} \rfloor + 1$

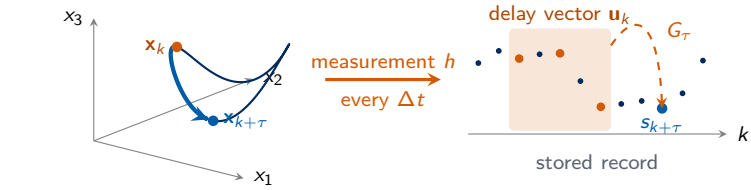
Limits of the Reconstruction



laser: field E, polarisation P, inversion W; $(E, P, W) \mapsto (-E, -P, W)$ leaves the Maxwell–Bloch equations and the intensity E^2 unchanged

- intensity: a non-generic measurement; symmetric states give one point, the sign of E is never recovered
- noise blurs nearby delay vectors; the theorem says nothing about noise
- contracting exponents, and with them D_{KY}, are poorly determined from a reconstruction

Forecast Map from Delay Vectors



$\mathbf{x}_k$, $\mathbf{x}_{k+\tau}$: hidden states; dots: stored record; orange dots: entries of the delay vector $\mathbf{u}_k$ ($m = 3$); blue dot: target $s_{k+\tau}$; dashed arrow: forecast map G_τ

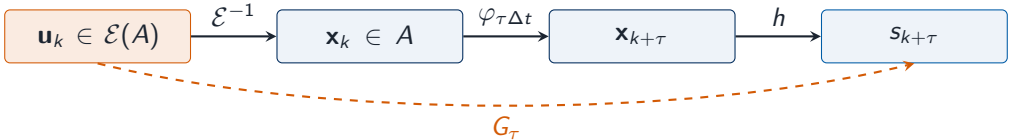
- G_τ goes from the delay vector directly to the future value, without the hidden phase space
- estimated by a model $\hat{G}_\tau$ from the pairs $(\mathbf{u}_k, s_{k+\tau})$ of the record; window $L = (m-1)\kappa + 1$

Forecast Map

Forecast map on the reconstructed attractor

$$G_\tau : \mathcal{E}(A) \to \mathbb{R}, \qquad G_\tau := h \circ \varphi_{\tau\Delta t} \circ \mathcal{E}^{-1}, \qquad G_\tau(\mathbf{u}_k) = s_{k+\tau}$$

read right to left: $\mathcal{E}^{-1}$ recovers $\mathbf{x}_k$ from $\mathbf{u}_k$; $\varphi_{\tau\Delta t}$, flow map of $\mathbf{F}$ over $\tau\Delta t$, gives $\mathbf{x}_{k+\tau}$; h measures it



- defined on admissible, noise-free delay vectors, the points of $\mathcal{E}(A)$; $\mathcal{E}^{-1}$ exists only where $\mathcal{E}$ is one-to-one
- laser: $\pm E$ not separated, yet G_τ exists, since symmetric states have the same future intensity

4

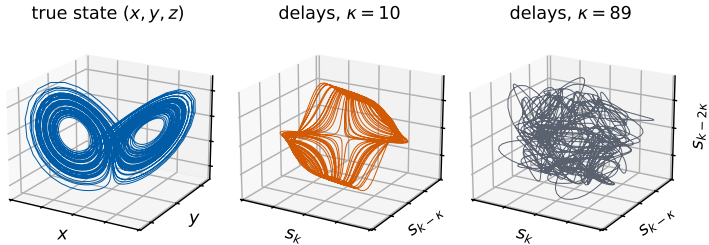
Delay and Dimension from the Record



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Choosing the Delay κ

Aim: the spacing κ of the delay vector $\mathbf{u}_k = (s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa})$, from the record alone.

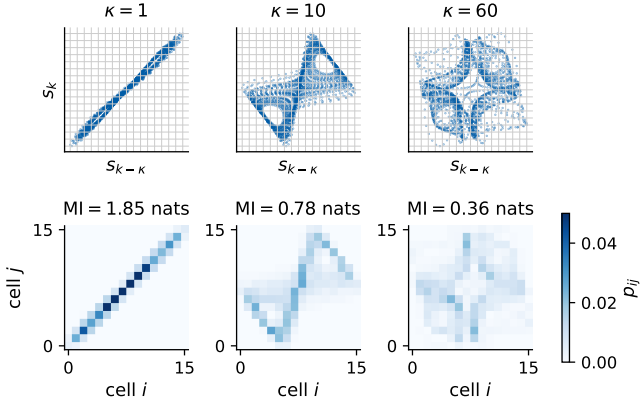


Lorenz-63: true state (x, y, z) and delay vectors $(s_k, s_{k-\kappa}, s_{k-2\kappa})$, $s_k = x(k\Delta t)$

- κ too small: $s_{k-\kappa} \approx s_k$, all points near the diagonal
- $\kappa = 10$: the two lobes reappear, smoothly deformed
- $\kappa = 89$, first zero of the autocorrelation function (ACF): the shape is lost

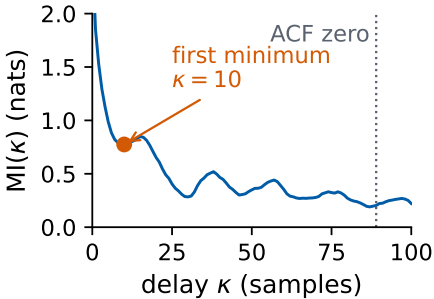
Mutual Information

Aim: a number that picks κ, the mutual information $\mathrm{MI}(\kappa)$ of $s_{k-\kappa}$ and s_k.



Lorenz x; top: pairs $(s_{k-\kappa}, s_k)$ for $\kappa = 1, 10, 60$ with the 16×16 grid; bottom: fraction p_{ij} of pairs in cell (i, j), shade of blue; $\mathrm{MI}(\kappa)$ above each panel, in nats

Mutual Information Against the Delay



$\mathrm{MI}(\kappa)$ of Lorenz x in nats against κ in samples; dot: first local minimum $\kappa = 10$; dotted: first zero of the ACF, 89

A. M. Fraser, H. L. Swinney, Phys. Rev. A 33, 1134 (1986)

Mutual information from the pairs $(s_{k-\kappa}, s_k)$

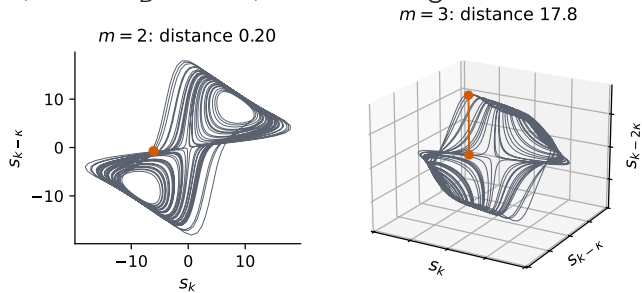
$$\mathrm{MI}(\kappa) = \sum_{i,j} p_{ij} \ln \frac{p_{ij}}{p_i p_j}$$

p_{ij}: fraction of pairs in cell (i, j); $p_i = \sum_j p_{ij}$, $p_j = \sum_i p_{ij}$

- $\mathrm{MI} = 0$ if and only if independent; zero ACF does not mean independence
- chosen κ: first local minimum, a heuristic

False Nearest Neighbours (FNN)

Aim: dimension m; $m > 2D_B$ suffices, but a record gives no reliable D_B or D_{KY}.



Lorenz x, $\kappa = 10$; left $(s_k, s_{k-\kappa})$, $m = 2$; right with $s_{k-2\kappa}$ added, $m = 3$; orange: one pair of nearest neighbours in $m = 2$

- close in $m = 2$, far in $m = 3$: a **false neighbour**; different states, different futures

M. B. Kennel, R. Brown, H. D. I. Abarbanel, Phys. Rev. A 45, 3403 (1992)

Determining m with the FNN Criterion

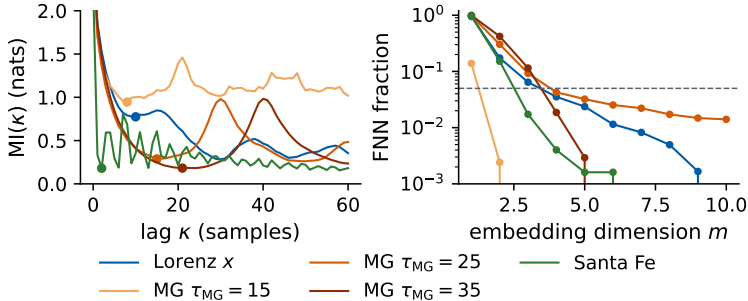
Nearest neighbour in m delay coordinates

$$r_k = \|\mathbf{u}_k^{(m)} - \mathbf{u}_j^{(m)}\| = \Big(\sum_{i=0}^{m-1} (s_{k-i\kappa} - s_{j-i\kappa})^2 \Big)^{1/2}, \qquad \Delta_k = |s_{k-m\kappa} - s_{j-m\kappa}|$$

κ from MI; $\mathbf{u}_j^{(m)}$: nearest neighbour of $\mathbf{u}_k^{(m)}$ among $|k-j| > 50$, another pass of the trajectory; Δ_k: separation in the added coordinate; σ_s: standard deviation of the record

1. **false neighbour** if $\Delta_k > 15\, r_k$ (ratio test; Kennel *et al.* add an absolute test, $\sqrt{r_k^2 + \Delta_k^2} > 2\sigma_s$)
2. $\mathrm{FNN}(m)$: fraction of k with a false neighbour
3. $m_{\min}$: first m with $\mathrm{FNN}(m) < 5\,\%$, an empirical m, not a proof of embedding

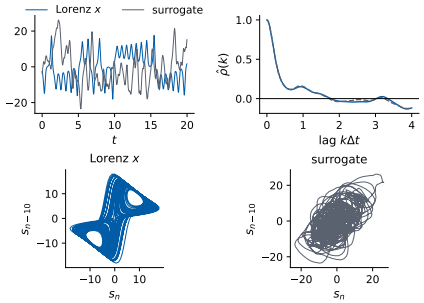
Dimension from the FNN Fraction



first 2500 samples (training block) only; left: $\mathrm{MI}(\kappa)$, dots at the chosen κ; right: fraction of false neighbours against m, dashed: 5 %

- m_{min}: first m below 5 %: Lorenz 4, MG 15 2, Santa Fe 3
- window $L = (m_{\mathrm{min}} - 1)\kappa + 1$

Same Correlations, Different Dynamics



top left: Lorenz x (blue) and its phase-randomised surrogate (grey) against time t; top right: their ACFs $\hat{\rho}$ against the lag; bottom: s_n against s_{n-10}

- surrogate: Fourier amplitudes of the record, random phases
- same ACF and spectrum; delay plot: folded attractor against a structureless cloud

J. Theiler *et al.*, Physica D 58, 77 (1992)

Phase Randomisation Keeps the Correlations

Finite Wiener–Khinchin theorem

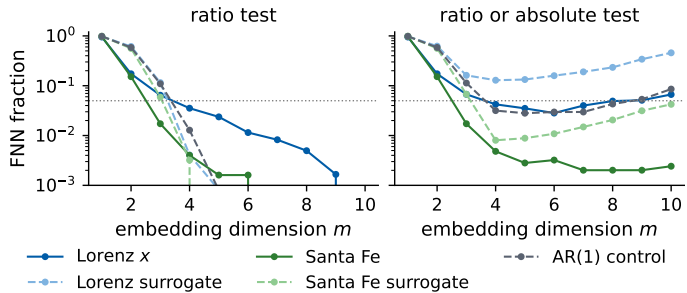
$$X_q = \sum_{k=0}^{N-1} z_k \, e^{-2\pi i q k/N}, \qquad R_c(n) := \frac{1}{N}\sum_{k=0}^{N-1} z_k z_{(k+n) \bmod N} = \frac{1}{N^2}\sum_{q=0}^{N-1} |X_q|^2 e^{-2\pi i q n/N}$$

$z_k = s_k - \bar{s}$; X_q: discrete Fourier transform; $R_c(n)$: circular autocovariance at lag n

- surrogate $\tilde{X}_q = |X_q| \, e^{i\theta_q}$, random $\theta_q = -\theta_{N-q}$, $\theta_0 = \theta_{N/2} = 0$: same $|X_q|$, same $R_c(n)$ at every lag
- Lorenz: circular ACFs agree to 2.2×10^{-16}, usual $\hat{\rho}$ to 0.013 up to lag 100
- iterative amplitude-adjusted Fourier transform (IAAFT): also the value distribution of the record, approximate spectrum

T. Schreiber, A. Schmitz, Phys. Rev. Lett. 77, 635 (1996)

False Neighbours of Chaos and Noise

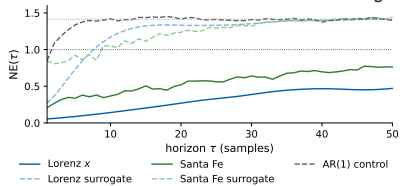


FNN fraction against m, first 2500 samples; solid: Lorenz, laser; dashed: their surrogates and AR(1), $s_k = 0.8\, s_{k-1} + \varepsilon_k$; left: ratio test; right: ratio or absolute test; dotted: 5 %

- $m = 2$: about $1/6$ false for chaos, over $1/2$ for noise
- $m = 4$: ratio test below 5 % for noise as well

Nearest-Neighbour Forecast

$$\hat{s}_{k+\tau} = s_{j+\tau}, \qquad \mathrm{NE}(\tau) = \frac{\left\langle \left(s_{k+\tau} - \hat{s}_{k+\tau}\right)^2 \right\rangle^{1/2}}{\sigma_s}$$

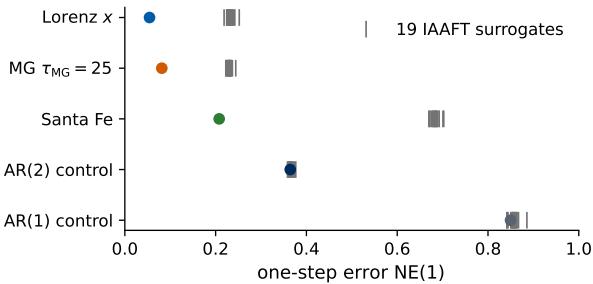


$\mathbf{u}_j$: nearest neighbour of the delay vector $\mathbf{u}_k$, $|k-j| > 50$; σ_s: standard deviation of the targets; NE: normalised error; dotted: forecast by the mean, $\mathrm{NE} = 1$, and by an unrelated value, $\mathrm{NE} = \sqrt{2}$

- NE(1): Lorenz 0.054, surrogate 0.27; laser 0.208, surrogate 0.85; AR(1) $\sqrt{2(1-0.8^2)} = 0.85$

E. N. Lorenz, J. Atmos. Sci. 26, 636 (1969)

Rank Test



dot: $\mathrm{NE}(1)$ of the record; ticks: 19 IAAFT surrogates; MG 25: Mackey–Glass, $\tau_{\mathrm{MG}} = 25$; AR(2): $s_k = 1.6\, s_{k-1} - 0.9\, s_{k-2} + \varepsilon_k$

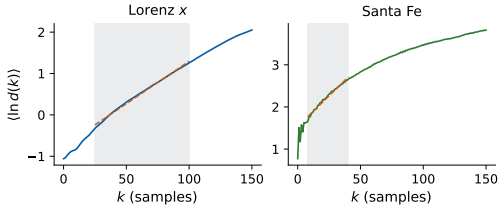
H_0: linear Gaussian process, seen through a fixed monotone function $\tilde{h}$

$$P(\text{rank of } Q_0 = 1 \,|\, H_0) = \frac{1}{M+1}$$

Q_0: statistic of the record, here $\mathrm{NE}(1)$; M surrogates; exchangeable under H_0

- $M = 19$: level 0.05
- rejected: Lorenz, MG 25, laser; kept: AR(1), AR(2)

Leading Exponent from Pairs of Neighbours



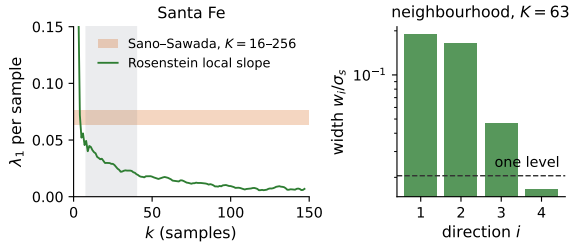
$\langle \ln \delta(n) \rangle$, written $\langle \ln d(k) \rangle$: mean log distance of neighbouring delay vectors after n steps; left: Lorenz; right: laser; shaded: fit range; c_l: start separation along the stretching directions with growth factors σ_l

$$\ln \delta(n) = \ln |c_1| + \ln \sigma_1(n\Delta t) + \tfrac{1}{2} \ln \left(1 + \sum_{l \geq 2} \frac{c_l^2}{c_1^2} \frac{\sigma_l^2}{\sigma_1^2}\right)$$

- $\langle \ln \sigma_1(t) \rangle / t \geq \lambda_1$: local slope falls towards $\lambda_1 \Delta t$ from above, then saturation

M. T. Rosenstein *et al.*, Physica D 65, 117 (1993)

Two Estimators on a Thin Attractor



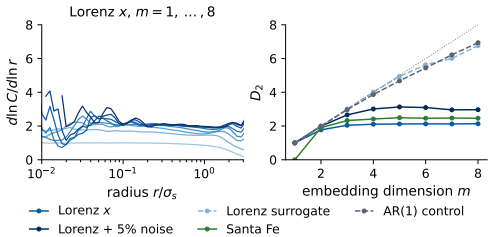
left: local slope of the laser curve, orange band: Sano–Sawada λ_1 for $K = 16$–256 neighbours; Sano–Sawada: local linear maps fitted to K nearest neighbours; right: root-mean-square widths w_i of $K = 63$ neighbourhoods, $m = 4$, dashed: one digitiser level

- Rosenstein: no range of constant slope, fit 8–40 gives 0.029 per sample
- Sano–Sawada: 0.064–0.076 per sample; thin directions hold only rounding

M. Sano, Y. Sawada, Phys. Rev. Lett. 55, 1082 (1985)

Correlation Dimension

$$\mathcal{C}_m(r) \coloneqq \frac{1}{N_\mathrm{c}} \sum_{j-i>50} \mathbf{1}\left(\|\mathbf{u}_i^{(m)} - \mathbf{u}_j^{(m)}\| < r\right) \propto r^{D_2}$$



$\mathcal{C}_m(r)$: fraction of the N_c pairs of delay vectors closer than the radius r; $\mathbf{1}$: 1 if true, else 0; left: local slopes of Lorenz, $m = 1$–8; right: D_2 against m, dotted: $D_2 = m$

- saturates: Lorenz ≈ 2.13 ($D_\mathrm{KY} = 2.06$), laser ≈ 2.46; noise grows with m: 6.76 and 6.94 at $m = 8$

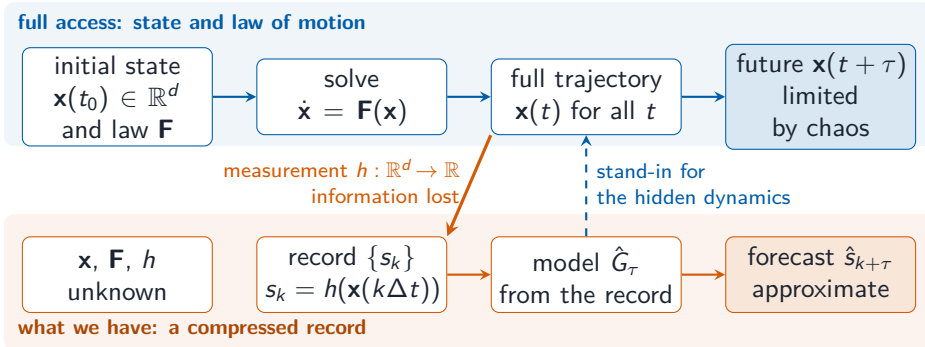
P. Grassberger, I. Procaccia, Phys. Rev. Lett. 50, 346 (1983)

Limits of the Diagnostics

- measurement noise fills small scales: plateau of D_2 for Lorenz $2.13 \to 2.97$ with $5\,\%$ noise
- rejected null: linear Gaussian process through a static measurement; nonlinear noise or a drifting record also reject it
- finite record: $D_2 \lesssim 2 \log_{10} N = 7.2$; noise records reach about 7 at $m = 8$

J.-P. Eckmann, D. Ruelle, Physica D 56, 185 (1992)

Full Access and the Compressed Record



blue row: forecast from state and law; orange row: forecast from the record alone

- a delay vector $\mathbf{u}_k$ replaces the unknown state (Takens, $m > 2D_B$)
- to build and judge $\hat{G}_\tau$: κ, m, D_{KY} and λ_1, all from the record

Record Quantities for a Forecast

needed for the forecast	obtained from	result
the state, from past values	$\mathrm{MI}(\kappa)$, $\mathrm{FNN}(m)$	delay vector $\mathbf{u}_k = (s_k, \ldots, s_{k-(m-1)\kappa})$
size of the attractor	Lyapunov spectrum	D_{KY}; check $m \geq \lceil D_{\mathrm{KY}} \rceil$
how far ahead	λ_1	$T_{\mathrm{pred}} = T_\lambda \ln(\Delta/\delta_0)$

- if $\mathbf{u}_k$ fixes $\mathbf{x}_k$, a forecast map $G_\tau(\mathbf{u}_k) = s_{k+\tau}$ exists; FNN supports it below the Takens bound
- chaos: G_τ is useful only up to about T_{pred}, for any model
- next: models $\hat{G}_\tau$ of G_τ learned from the record, from linear models to neural networks

Characteristics of the Signals

signal	κ	m	$L = (m-1)\kappa + 1$	D_{KY}	λ_1 per sample	T_λ (samples)
Lorenz x	10	4	31	2.06	0.0181	55
MG 15	8	2	9	1.00	≈ 0	∞
MG 25	15	4	46	3.16	0.0099	101
MG 35	21	4	64	3.41	0.0029	340
Santa Fe	2	3	5	2.05 / 2.16	0.029	34

κ from MI, m from FNN; D_{KY}, λ_1: models from the equations, laser from the record (λ_1: Rosenstein; D_{KY}: Sano–Sawada, $m = 3\,/\,4$); κ, L, $T_\lambda = 1/\lambda_1$ in samples

- $\lceil D_{\mathrm{KY}} \rceil \leq m$ for every signal; chaotic signals: FNN gives $m < 2D_{\mathrm{KY}}$
- horizons $\tau \leq 50$: about one T_λ for Lorenz; 1.5–3.3 T_λ for Santa Fe; below one for MG 25, 35
- MI, FNN: a heuristic window; longer windows chosen on validation data

Summary

- task: forecast or recall $s_{k\pm\tau}$ from the record; forecasts scored by $C(\tau)$
- the state $\mathbf{x}$ lives on an attractor; chaos: errors grow as $e^{\lambda_1 t}$; horizon $T_{\mathrm{pred}} = T_\lambda \ln(\Delta/\delta_0)$ (Lorenz-63: $T_{\mathrm{pred}} = 20.3$, $T_\lambda = 1.1$ time units)
- Takens: for $m > 2D_B$ the delay map is one-to-one for almost every measurement; κ from MI, m from FNN on the training block, below $2D_{\mathrm{KY}}$ here, so the existence of the forecast map G_τ is an assumption supported by FNN
- chaos or noise: nearest-neighbour forecast, rank test against surrogates, D_2 saturating with m; laser: structure beyond linear noise, $D_2 \approx 2.46$, $\lambda_1 > 0$ by both estimators

Thank you for your attention

5
Extras



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Derivation of Liouville's Theorem

$$\frac{\mathrm{d}}{\mathrm{d}t} M_t = J(\varphi_t(\mathbf{x}_0))\, M_t, \qquad M_0 = I \qquad \text{differentiate } \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) \text{ by } \mathbf{x}_0$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \det M = \det M \, \operatorname{tr}\Big(M^{-1} \frac{\mathrm{d}M}{\mathrm{d}t}\Big) \qquad \text{Jacobi's formula}$$

$$\operatorname{tr}\big(M_t^{-1} J M_t\big) = \operatorname{tr} J = \nabla \cdot \mathbf{F} \qquad \text{cyclic trace}$$

$$\Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t} \det M_t = \big(\nabla \cdot \mathbf{F}\big)\big(\varphi_t(\mathbf{x}_0)\big)\, \det M_t$$

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \int_{A_0} \big(\nabla \cdot \mathbf{F}\big)\big(\varphi_t(\mathbf{x}_0)\big) \det M_t \; d^d x_0 \qquad \text{from } V(t) = \textstyle\int_{A_0} \det M_t \, d^d x_0$$

$$= \int_{A_t} \nabla \cdot \mathbf{F} \; d^d y \qquad \mathbf{y} = \varphi_t(\mathbf{x}_0)$$

$M_t = M_t(\mathbf{x}_0)$: Jacobian of the flow map; $J_{ij} \coloneqq \partial F_i / \partial x_j$: Jacobian of $\mathbf{F}$; Jacobi's formula holds for any invertible matrix $M(t)$

Stochastic Processes

A stochastic process gets a new random number ε_k at every step; nobody can know it in advance.

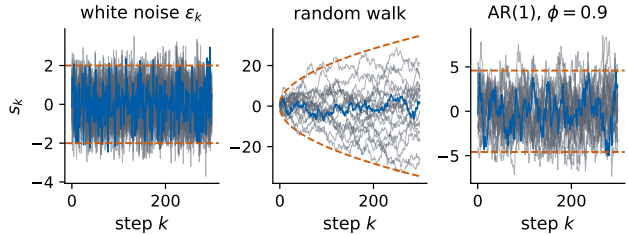
Three basic processes

white noise:	$s_k = \varepsilon_k$	a new random number
random walk:	$s_k = s_{k-1} + \varepsilon_k$	last value + kick
AR(1), autoregressive of order 1:	$s_k = \phi\, s_{k-1} + \varepsilon_k$	fraction ϕ of last value + kick

ε_k: independent random numbers, mean 0, variance σ^2; ϕ, $|\phi| < 1$: memory coefficient; $\phi = 0$ gives white noise, $\phi = 1$ the random walk

- **stationary**: mean and correlations $\mathrm{cov}(s_k, s_{k+n})$ do not depend on k
- white noise and AR(1) are stationary; the random walk is not, its variance grows as $k\sigma^2$

Realisations and the AR(1) Process

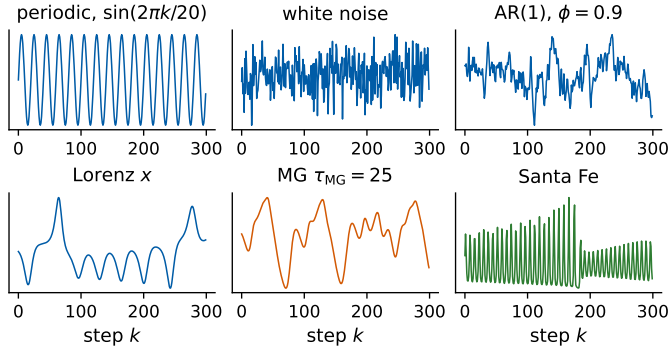


20 realisations (grey, one blue), $\sigma = 1$, $\phi = 0.9$; dashed: ± 2 standard deviations of the process

- AR(1), $s_k = \phi\, s_{k-1} + \varepsilon_k$: $\rho(n) = \phi^n$, memory $-1/\ln\phi \approx 9.5$ steps
- best forecasts: mean (white noise), last value (random walk), $\phi^\tau s_k$ (AR(1))
- forecast error: from future kicks ε (noise), from the initial error (chaos)

G. U. Yule, Phil. Trans. R. Soc. A 226, 267 (1927)

Six Test Signals



First 300 samples of each signal

- known type: periodic $s_k = \sin(2\pi k/20)$; white noise; AR(1) with $\phi = 0.9$
- chaotic: Lorenz-63 x, Mackey–Glass $\tau_{\mathrm{MG}} = 25$, Santa Fe laser

Autocorrelation Function (ACF)

Exact definitions: infinite stationary record $\vec{s} = \{s_1, s_2, s_3, \dots\}$, no end

$$\text{mean:} \quad \bar{s} \coloneqq \lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} s_k$$

$$\text{autocovariance at lag } n\text{:} \quad R(n) \coloneqq \lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} (s_k - \bar{s})(s_{k+n} - \bar{s})$$

$$\text{autocorrelation at lag } n\text{:} \quad \rho(n) \coloneqq R(n)/R(0)$$

n: lag in samples; $R(0)$: variance of $\vec{s}$; $N \to \infty$: never available in practice

- $\rho(n)$: correlation of s_k with s_{k+n}; $\rho(0) = 1$; $\rho(n) \approx 1$: s_{k+n} follows s_k; ≈ 0: no linear relation; < 0: opposite sign
- only linear dependence: chaos and noise can share the same $\rho(n)$

G. E. P. Box *et al.*, *Time Series Analysis: Forecasting and Control*, Wiley (2015)

ACF Estimate from a Finite Record

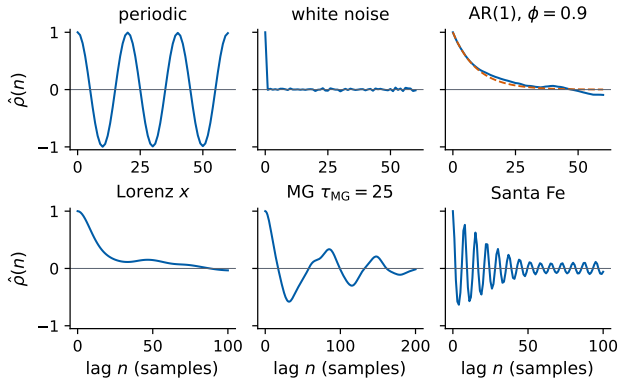
Estimate: finite record $\vec{s} = \{s_1, s_2, \ldots, s_N\}$, the N stored samples

$$\bar{s} := \frac{1}{N}\sum_{k=1}^{N} s_k, \qquad \hat{\rho}(n) := \frac{\sum_{k=1}^{N-n}(s_k - \bar{s})(s_{k+n} - \bar{s})}{\sum_{k=1}^{N}(s_k - \bar{s})^2}$$

sums stop at N, the record mean $\bar{s}$ of the N samples estimates the process mean; $\hat{\rho}(n)$ fluctuates around $\rho(n)$, $\hat{\rho}(n) \to \rho(n)$ as $N \to \infty$

- computed from the stored samples only
- large lags: numerator of only $N - n$ terms, $\hat{\rho}(n)$ biased towards zero

ACF of Known and Chaotic Signals



blue: $\hat{\rho}(n)$ from $N = 4000$ samples against lag n; orange dashed: exact $\rho(n) = \phi^n$ of AR(1), $\phi = 0.9$; periodic signal: $\sin(2\pi k / T_{\mathrm{p}})$, $T_{\mathrm{p}} = 20$

- periodic: back to ≈ 1 every period; noise and chaos: decay

Power Spectral Density (PSD)

PSD of an infinite stationary record $\vec{s} = \{s_1, s_2, \dots\}$

mean: $\bar{s} := \lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} s_k$

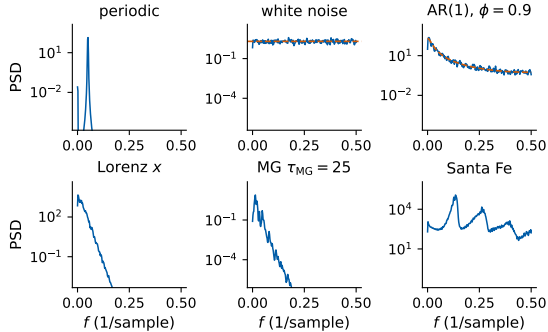
autocovariance at lag n:
$R(n) := \lim_{N\to\infty} \frac{1}{N} \sum_{k=1}^{N} (s_k - \bar{s})(s_{k+n} - \bar{s})$,
$R(-n) = R(n)$

power spectral density, $|f| \le \frac{1}{2\Delta t}$:
$S(f) := \Delta t \sum_{n=-\infty}^{\infty} R(n)\, e^{-2\pi i f n \Delta t}$

$S(f)\,\mathrm{d}f$: variance carried by frequencies in $[f, f + \mathrm{d}f]$;
$\int_{-1/(2\Delta t)}^{1/(2\Delta t)} S\,\mathrm{d}f = R(0)$; one-sided estimates give $2S(f)$, $f > 0$

- peak at f: oscillation with period $1/f$
- $R(n)$ and $S(f)$ carry the same information (Wiener–Khinchin)

PSD of Known and Chaotic Signals



estimated $S(f)$ against frequency f per sample, logarithmic vertical axis

- periodic: one line at $f = 1/T_{\mathrm{p}}$; white noise flat; AR(1) broadband
- Lorenz x, Mackey–Glass broadband, Santa Fe peaks at $f_{\mathrm{a}} \approx 0.138$ and $f_{\mathrm{b}} \approx 0.262$, near $2f_{\mathrm{a}}$ without coinciding: the PSD alone does not separate chaos from noise

Spectrum from a Finite Record

Estimates of $S(f)$ from a finite record $\vec{s} = \{s_1, \ldots, s_N\}$

$$\text{periodogram:} \quad \hat{S}(f_q) \coloneqq \frac{2\Delta t}{N} \left| \sum_{k=1}^{N} (s_k - \bar{s})\, e^{-2\pi i\, qk/N} \right|^2, \qquad f_q \coloneqq \frac{q}{N\Delta t}$$

$$\text{Welch average:} \quad \hat{S}_{\mathrm{W}}(f) \coloneqq \frac{1}{M} \sum_{r=1}^{M} \hat{S}^{(r)}(f)$$

$\bar{s}$: record mean; $q = 1, \ldots, N/2$; $\hat{S}^{(r)}$: periodogram of segment r; the M segments overlap by half

- $\hat{S}$ from data, S from the infinite record; $\hat{S} \to 2S$ on average, $f > 0$
- one periodogram scatters strongly; the average of M is smooth, on a coarser frequency grid

P. D. Welch, IEEE Trans. Audio Electroacoust. 15, 70 (1967)

Backpropagation Through Time

The recurrent weights act at every step of the unrolled network, so their gradient collects one contribution from every step. The chain rule, run backwards from the last step, gives

$$\frac{\partial \mathcal{L}}{\partial W_h} = \sum_{j=1}^{k} \frac{\partial \mathcal{L}}{\partial \mathbf{h}_k} \frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_j} \frac{\partial^+ \mathbf{h}_j}{\partial W_h}.$$

- this is backpropagation through time (BPTT), and $\partial^+ \mathbf{h}_j / \partial W_h$ is the direct effect of W_h at step j
- an input $k - j$ steps in the past reaches the loss only through the factor $\partial \mathbf{h}_k / \partial \mathbf{h}_j$

P. J. Werbos, Proc. IEEE 78, 1550 (1990), doi:10.1109/5.58337; Y. Bengio, P. Simard, P. Frasconi, IEEE Trans. Neural Netw. 5, 157 (1994), doi:10.1109/72.279181

Vanishing and Exploding Gradients

Each step of the cell contributes one Jacobian to the path from $\mathbf{h}_j$ to $\mathbf{h}_k$,

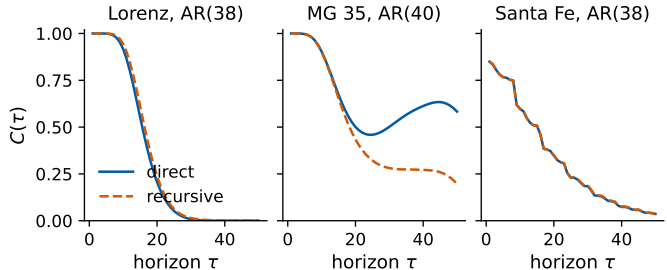
$$\frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_j} = D_k W_h D_{k-1} W_h \cdots D_{j+1} W_h, \qquad D_i \coloneqq \operatorname{diag}(1 - \mathbf{h}_i^2), \qquad \left\| \frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_j} \right\|_2 \approx e^{\gamma (k-j)}$$

D_i: slopes of tanh at step i, between 0 and 1; γ: average growth rate per step

- a product of $k - j$ similar factors grows or decays exponentially
- $\gamma < 0$: the gradient vanishes, distant inputs drop out of the training signal
- $\gamma > 0$: the gradient explodes, rare very large updates spoil the weights

Y. Bengio, P. Simard, P. Frasconi, IEEE Trans. Neural Netw. 5, 157 (1994), doi:10.1109/72.279181;
R. Pascanu, T. Mikolov, Y. Bengio, ICML 2013, PMLR 28, 1310, arXiv:1211.5063

Direct and Recursive Linear Forecasts



$C(\tau)$ on the test block; direct: one AR map per horizon, $\hat{s}_{k+\tau} = c_\tau + \sum_i w_{\tau,i} s_{k-i}$; recursive: one one-step AR model, each forecast $\hat{s}_{k+1}$ fed back as the newest input until $\hat{s}_{k+\tau}$

- an iterated linear model relaxes to its mean
- direct wins on Mackey–Glass, on Lorenz and Santa Fe both lose the signal within about 20 steps

Backshift Operator

A compact notation for shifts in time makes long models readable.

$$Bs_k := s_{k-1}, \qquad B^2 s_k = s_{k-2}, \qquad (1 - B)\, s_k = s_k - s_{k-1}$$

- AR(2), $s_k = \phi_1 s_{k-1} + \phi_2 s_{k-2} + \varepsilon_k$, becomes $(1 - \phi_1 B - \phi_2 B^2)\, s_k = \varepsilon_k$
- in general we collect the weights into two polynomials in B,

$$\Phi_p(B) := 1 - \phi_1 B - \cdots - \phi_p B^p, \qquad \Theta_q(B) := 1 + \theta_1 B + \cdots + \theta_q B^q$$

ARIMA

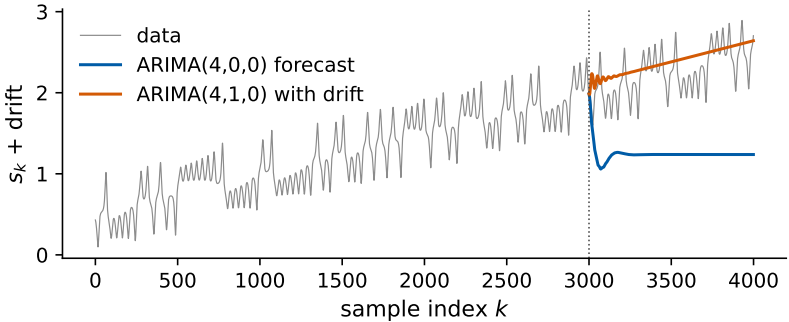
Autoregressive (AR), moving-average (MA) and ARMA models assume a fixed mean. A signal with a trend has none, but its differences do. ARIMA, the autoregressive integrated moving average, applies ARMA to the differences.

$$\Delta s_k := s_k - s_{k-1}, \qquad \Delta^d s_k := \Delta(\Delta^{d-1} s_k), \qquad z_k := \Delta^d s_k$$

1. difference the signal d times
2. forecast the differences with ARMA(p, q), $\hat{z}_{k+1} = c + \sum_{i=1}^{p} \phi_i z_{k+1-i} + \sum_{j=1}^{q} \theta_j e_{k+1-j}$
3. add back, $\hat{s}_{k+1} = s_k + \hat{z}_{k+1}$ for $d = 1$, $\hat{s}_{k+1} = 2s_k - s_{k-1} + \hat{z}_{k+1}$ for $d = 2$

The trainable parameters are those of ARMA, and d is chosen by us. G. E. P. Box, G. M. Jenkins, G. C. Reinsel, G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Wiley (2015)

ARIMA on a Drifting Signal



Lorenz x plus a linear drift; each model forecasts the whole test block from its first step.

- without differencing ($d = 0$) the forecast relaxes to the mean
- with $d = 1$ it continues the trend, but neither follows the oscillation